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FRONTIERS IN CARDIOVASCULAR QUALITY AND OUTCOMES

Sports and Exercise Cardiology: A New Frontier in Cardiovascular Disease Management

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ABSTRACT: Exercise and physical activity are recognized as essential tools to optimize fitness and longevity, improve overall health, and reduce the risk of cardiovascular disease and other major comorbidities. However, there are inherent risks associated with participation in sports and exercise, regardless of the level of competition or skill required. As such, there is a need for cardiologists to understand the unique aspects of competitive and recreational athletics, as well as sport and exercise science, as it relates to overall heart health and cardiovascular risk. In this article, we provide an overview of the essential aspects of sports and exercise cardiology. We describe the roles and responsibilities of a sports and exercise cardiologist as a member of a comprehensive team of experts who provide care of athletes at all levels of sport, emphasize the importance of risk stratification and prevention of sudden cardiac arrest, the role of exercise for maintaining and optimizing overall health in the general population, describe the growing field of sports electrophysiology and also discuss unique aspects of sports cardiology applied to environmental health and care of the tactical athlete. Knowledge gaps and directions for future clinical and research investigation are reviewed.

Key Words: cardiovascular disease ■ death, sudden, cardiac ■ electrophysiology ■ exercise ■ longevity ■ risk factors ■ sports

In recent years, there has been increased attention placed on the role of the sports and exercise cardiologist in the management of professional and recreational athletes.[1] There is a complex interplay between exercise and cardiovascular disease (CVD) among individuals at all levels of sport. For example, it is well established that routine adherence to an exercise program reduces the long-term risk of cardiometabolic disease, ischemic heart disease, and cerebrovascular disease,[2] but acutely, exercise may actually increase the risk of adverse events such as sudden cardiac arrest (SCA) or stroke.[3–5] In addition, cardiac structure and function adapt over time in response to routine exercise through a process referred to as exercise-induced cardiac remodeling (EICR).[1] EICR may take on an appearance of cardiomyopathy and increase the risk of arrhythmias. As such, medical providers must be able to differentiate an adaptive response to exercise from a potentially life-threatening cardiomyopathy.

Athletes are frequently referred for cardiac evaluation for additional reasons, including identification of factors limiting exercise capacity and, in some cases, to establish formal exercise training zones to facilitate overall health, achieve personal fitness goals, or improve athletic performance. In addition, there has been increased attention on cardiovascular health among tactical athletes—individuals who use training and athleticism for service to the community (eg, firefighters, police, military) as opposed to competition or performance. As such, there is a demand for cardiologists with expertise in sports and exercise science who can work with athletes and the community to optimize health and reduce risk of adverse events across all levels of competition or sport. This review provides a concise yet comprehensive overview of the emerging specialty of sports and exercise cardiology and practical applications of sports and exercise science applied to athletic populations. Key knowledge gaps and future directions are emphasized with the goal of directing future research to further optimize care among individuals participating in sports and exercise across the spectrum of competition.

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Nonstandard Abbreviations and Acronyms

ACM	arrhythmogenic cardiomyopathy
AED	automated external defibrillator
AF	atrial fibrillation
CAC	coronary artery calcium
CPET	cardiopulmonary exercise test
CPR	cardiopulmonary resuscitation
CR	cardiac rehabilitation
CVD	cardiovascular disease
EAP	emergency action plan
EICR	exercise-induced cardiac remodeling
HR	heart rate
LVEF	left ventricular ejection fraction
MSS	maximal steady-state
PFA	pulse field ablation
PPCS	preparticipation cardiovascular screening
PREVENT	Predicting Risk of CVD Events
RR	risk ratio
RTP	return-to-play
SCA	sudden cardiac arrest
SCD	sudden cardiac death

THE SPORTS CARDIOLOGY TEAM: ROLES AND RESPONSIBILITIES

Professional and recreational athletes receive care from multidisciplinary teams comprised of athletic trainers, physical therapists, orthopedic surgeons, and primary care physicians. Cardiologists provide additional expertise not provided by those specialists. While the American Board of Internal Medicine does not have an official competency statement regarding core competencies, there is a basic fund of knowledge necessary for the cardiologists engaging in this discipline.[1] Specifically, several areas related to sports and exercise cardiology are essential for cardiovascular care of the athlete[1]:

1. Management of CVD among athletes.
2. Differentiation of EICR from acquired cardiomyopathies.
3. Evaluation of factors limiting exercise performance and determining the cause of symptoms.
4. Preparticipation screening and reducing the risk of SCA.
5. Understanding the impact of environmental stressors on athletic performance.

Although this list is not exhaustive, it demonstrates the key components forming the framework for sports and exercise cardiology. Although athletes may be managed by athletic trainers and exercise physiologists, referral to a sports and exercise cardiologist may be necessary for several reasons. Table 1 provides a comprehensive list of the roles and responsibilities of sports and exercise cardiology as it relates to care of the athlete, regardless of the level of competition, as well as indications for referral to a sports and exercise cardiologist.

Comprehensive cardiovascular care of athletes is achieved by a team of cardiologists with expertise in the following areas: sports and exercise physiology, electrophysiology, heart failure and cardiomyopathies, cardiovascular imaging, congenital cardiology, and interventional cardiology. These specialists are necessary to manage the multiple issues that arise in relation to issues encountered among athletes. Unique features of sports and exercise cardiology are displayed in Figure 1.

Sports and exercise cardiologists naturally serve as educators and advisors on individual and societal levels. At the patient level, sports cardiologists risk-stratify athletes for major adverse cardiovascular events, including SCA,[6] develop personalized plans for achieving fitness goals and improving athletic performance,[7] and identify factors limiting exercise capacity.[1] Community organizers and legislators on a local, regional, and national level rely on sports cardiologists for multiple issues related to societal health. SCA is a case in point. For example, the incidence of SCA during marathons has remained relatively stable over the past 2 decades, at ≈1 in 184 000 from 2000 to 2009 and ≈1 in 172 000 from 2010 to 2019.[4,8] However, there has been an ≈50% reduction in the incidence of sudden cardiac death (SCD) due to community education, increased awareness of cardiac risk factors and implementation of emergency action plans (EAP) at sporting events.[9] To date, over 40 states in the United States require training in cardiopulmonary resuscitation (CPR) for high school graduation. These

Table 1. Roles and Responsibilities of a Sports and Exercise Cardiologist as Well as Indications for Referral

Roles and responsibilities
Patient and community education
Local, regional, and national legislation
Preparticipation screening
Reduce the risk of sudden cardiac arrest and sudden cardiac death
Primary and secondary risk prevention of cardiovascular disease
Promote overall health and longevity
Indications for referral to sports and exercise cardiology
Distinguish exercise-induced cardiac remodeling from acquired cardiomyopathies
Performance optimization: establish training zones
Identify factors limiting exercise tolerance
Arrhythmia management
Support tactical athletes
Environmental medicine

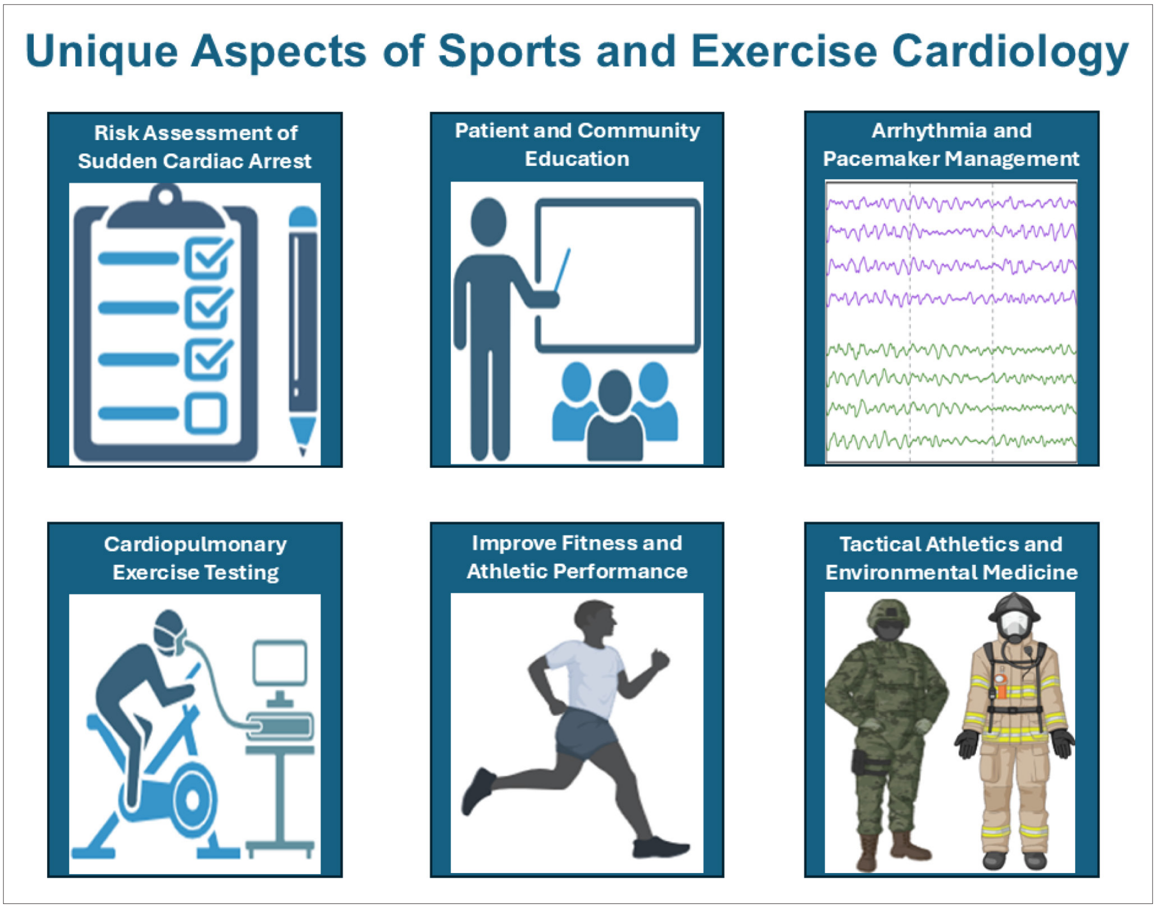

Figure 1. Key aspects of sports and exercise cardiology.
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initiatives exemplify the unique role that sports and exercise cardiologists have in shaping policy with the goal of improving health and reducing risk both on an individual and societal level.

RISK STRATIFICATION AND PREVENTION OF SCA

SCA and Sudden Cardiac Death

SCA/SCD is the leading medical cause of death among young competitive athletes.[10] The most common cause of SCA among athletes is sudden unexplained death with a structurally normal heart at autopsy.[10–12] However, heritable conditions are the most common identifiable cause of SCA among young athletes (those <35 years of age) and include abnormalities such as hypertrophic cardiomyopathy, congenital abnormalities, and coronary artery anomalies.[6,9,10,12,13]

Although a detailed analysis of all causes of SCA/SCD is beyond the scope of this review, there are a few points to be made. First, hypertrophic cardiomyopathy is the most common heritable cardiac disease, impacting ≈1 in 500 individuals,[14] and in many analyses, has been reported as the most common cause of SCD in young athletes.[15] Coronary artery anomalies have been reported as the second most common cause of SCD in young athletes, accounting for 17% of deaths.[15] However, in a case series of SCD in military recruits (N=126), coronary anomalies were the most common cause and accounted for 61% of cases.[16] Myocarditis is another major cause of SCD. In an analysis of autopsy records of Air Force recruits who died from SCD (19 cases among 1 606 167 recruits from 1965 to 1985), myocarditis was the most common cause.[17]

Arrhythmogenic cardiomyopathy (ACM) is another major cause of SCD.[18] Although ACM is classically described as a heritable cardiomyopathy resulting from mutations in desmosomal proteins,[19] exercise-induced ACM has been reported among genotype-negative athletes.[20] In an analysis of genotype-positive ACM and exercise-induced ACM athletes, over a 96-month period of follow-up, the incidence of ventricular arrhythmias was high in both populations (64% in genotype-positive ACM and 40% in exercise-induced ACM).[20] Thus, athletes with gentyope-negative, exercise-induced ACM should be followed closely over time to monitor for the presence of life-threatening arrhythmias.

Risk Stratification

Preparticipation cardiovascular screening (PPCS) is performed with a thorough history and physical examination, with or without the inclusion of a 12-lead ECG to diagnose conditions associated with SCA/SCD. Particular symptoms of concern to sports and exercise cardiologists that require additional testing include syncope, palpitations, angina, and exertional dyspnea, and detection of these symptoms among athletes may be improved by inclusion of educational videos during PPCS.[21] The inclusion of ECG as a component of PPCS among young competitive athletes is controversial.[22–24] PPCS with ECG is performed in some countries and sporting leagues, largely driven by the ability to improve the diagnostic accuracy beyond history and physical for conditions associated with SCA/SCD.[25–27] The European Society of Cardiology has supported widespread ECG screening based on observational outcomes data from Italy.[28,29] However, these data have not been replicated in other parts of the world.[30] Thus, widespread ECG screening of young competitive athletes has been traditionally opposed by cardiovascular societies in the United States,[22,24] given concerns for high false positive rates,[31] costs associated with screening and secondary testing,[32] the known harms associated with cardiovascular testing and interventions, inadequate expertise of providers interpreting athletic ECGs,[33,34] and absence of efficient and equitable access to secondary cardiovascular testing. In the most recent 2025 American College of Cardiology/American Heart Association scientific statement on Clinical Considerations for Competitive Sports Participation for Athletes with Cardiovascular Abnormalities,[35] the inclusion of ECG as a component of PPCS for young competitive athletes was deemed reasonable so long as there are available physicians trained in athletic ECG interpretation and equitable access to high-quality secondary testing for athletes with abnormal ECGs. Many US colleges and professional sporting leagues perform more extensive PPCS that includes ECG, and in some cases, cardiac imaging. That said, inclusion of cardiac imaging is not supported by data and is not recommended for widespread screening.[35]

Controversy will always exist regarding optimal PPCS strategies for young competitive athletes. However, it is incontrovertible that a prompt emergency medical response improves survival following SCA in both athletes and nonathletes.[9,36–38] An effective emergency medical response requires recognition of SCA, prompt bystander CPR, the availability of an automated external defibrillator (AED) with rapid application and defibrillation if indicated, and immediate contact to emergency medical services. An EAP is a document that outlines standard procedures in the case of a catastrophic event or injury, including SCA.[39] EAPs should be widely distributed and rehearsed at least annually. Recently, the National Football League Smart Hearts Coalition led national efforts to create legislation to mandate EAPs, CPR training and education for coaches, and access to AEDs within 1 to 3 minutes of sporting venues for high schools throughout the United States.[40] Efficient access to AEDs is essential as survival from exertional SCA is higher for events that occur during games or formal competition as opposed to routine practices where there are fewer bystanders and less access to AEDs.[9]

Although national initiatives continue to evolve, significant disparities exist.[41] In a 9-year study of SCA among young competitive athletes in the US, non-White athletes were more likely to die from SCA after controlling for other factors.[9] In another analysis, minority high school athletes had reduced survival following SCA compared with White non-Hispanic athletes, and there was a non-significant increase in survival found with increasing median household income and with decreasing percent minority students or the proportion of students on free or reduced-cost lunch.[18] In addition, disparities in outcomes exist between Black and White athletes without a clear explanation.[6] These findings suggest that social determinants of health may contribute to racial disparities in SCA outcomes.[6,41] Disparities also exist based on the level of competition. In a study of Amateur Athletic Union youth club basketball teams (N=449), only 6% had an EAP, 35% required CPR training for their coaches, and AEDs were only available in 45% and 35% of practices and games, respectively.[37] An additional study demonstrated that poor emergency preparedness may lead to worse outcomes as male adolescent basketball players with SCA were more likely to receive bystander CPR, AED use, and had higher survival at school-sponsored versus amateur athletic union-sponsored events.[42]

EXERCISE CONSIDERATIONS IN COMPETITIVE ATHLETES

Diagnostic Assessments

Several unique factors specific to athletes arise during clinical evaluations and diagnostic assessments. First, a 12-lead ECG often displays patterns suggestive of cardiovascular pathology. The first criteria for ECG interpretation of athletes were developed in 2005, and since that time, multiple revisions have been developed to assist clinicians in distinguishing adaptive patterns from true pathology.[28] Current criteria generally divide ECG changes into normal variants commonly observed in athletes, borderline findings that may require additional investigation, and abnormal findings requiring follow-up evaluation.[43]

EICR describes the structural, functional, and electrical remodeling that occurs in response to routine exercise and forms the basis for the structural and functional characteristics that typify the athletic heart.[2]

EICR varies considerably between sporting disciplines since acute hemodynamic responses and chronic adaptations vary in accordance with loading conditions associated with any specific sport. Recognizing this high degree of variability, exercise has historically been divided into 2 broad categories: Strength and endurance (Figure 2).[2,44] The strength component of exercise refers to the intensity of muscle contractions required to complete a task relative to the individual's maximum voluntary contraction.[2,44] Strength exercise introduces a pressure load on the heart and leads to concentric remodeling over time. The endurance component of exercise refers to the aerobic power required relative to maximum oxygen uptake (VO_{2max}).[2,44] Endurance exercise introduces a volume load on the heart and, over time, leads to eccentric remodeling.[2] However, in reality, the majority of exercises have varying degrees of strength and endurance exercise, with overlap in the degree of cardiac remodeling that occurs in response to the athlete's chosen sport, position, and volume of exercise.[2]

The ability to distinguish between EICR and cardiomyopathy is one of the most fundamental aspects of sports cardiology.[1] However, this task may be challenging since EICR leads to gray zones, whereby changes in ventricular structure may be similar in appearance to an overt cardiomyopathy, and the ability to distinguish the 2 is not intuitively obvious. Three general gray zones have been described.[1] First, the physiological concentric hypertrophy resulting from participation in high-strength forms of exercise may appear similar to morphological changes observed with hypertrophic cardiomyopathy or hypertensive heart disease. However, EICR generally leads to symmetrical hypertrophy with a ventricular wall thickness not exceeding 15 mm.[1,45] Second, physiological eccentric hypertrophy resulting from high-endurance forms of exercise may appear similar to a dilated cardiomyopathy. However, contractility is typically preserved in EICR with normal/low normal systolic function. Finally, endurance exercise may lead to RV chamber dilatation that may appear similar to arrhythmogenic ventricular cardiomyopathy. However, RV systolic function and systolic

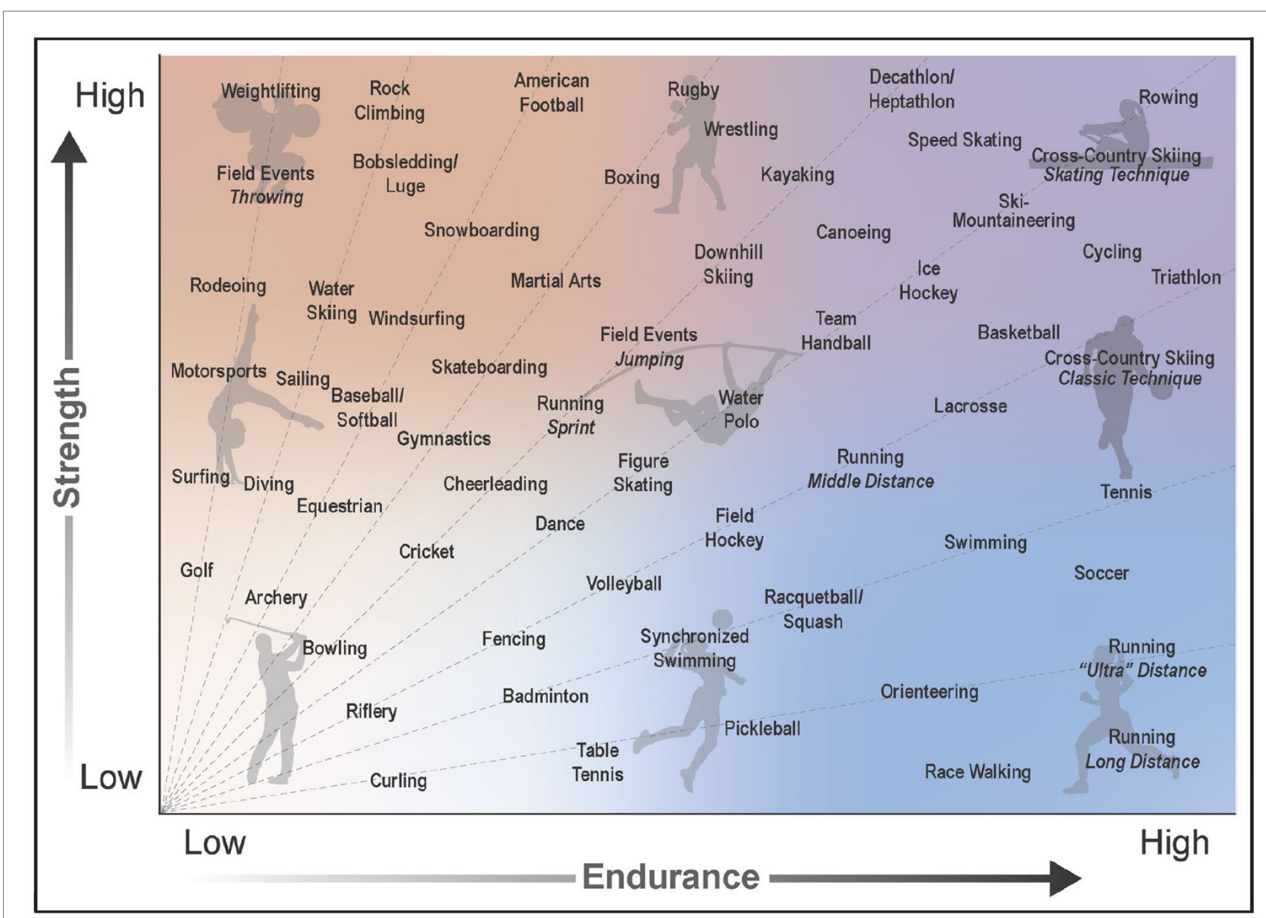

Figure 2. Classification scheme for competitive sports stratified according to relative strength and endurance intensities. Sports are displayed on arcs that signify the relative contributions of strength and endurance that accompany each specific sport. Actual exercise intensities may vary according to several factors, including playing position in certain team sports, event within an individual sport, varying training intensities according to level of competition and individual athlete preference, time of season or year, and environmental stressors. Reprinted from Kim et al[35] with permission. Copyright ©2025, American Heart Association Inc.

pressures are preserved among athletes with EICR. Differentiating between EICR and true cardiomyopathies requires a multifaceted approach, including imaging (echocardiography, cardiac magnetic resonance imaging) and functional assessments with cardiopulmonary exercise testing (CPET), as well as arrhythmia monitoring in select cases. However, a period of detraining may be necessary to monitor for signs of reverse remodeling with regression of the hypertrophy characteristic of EICR.[20,46]

Rarely, athletes engaged in dynamic exercise may have a reduced left ventricular ejection fraction (LVEF). In the Pro@Heart study, comprised of competitive, mostly male, endurance athletes, ≈10% had an LVEF <50% and reduced global longitudinal strain.[47] Interestingly, these athletes had a different genetic profile when compared with athletes in this same cohort who had a normal LVEF.[47] More specifically, athletes with reduced LVEF had an increased polygenic risk score when compared with those with normal LVEF, with genetic variants previously shown to increase risk of dilated cardiomyopathy in the general population.[47,48]

Training Zones and Performance-Based Assessments

Recreational and professional athletes routinely undergo performance-based assessments for several reasons, including establishing training zones, evaluating impairments in exercise capacity, and for return-to-play (RTP).[1,7] Training zones may be used by professional or recreational athletes for a variety of reasons and are attractive, since they allow for the implementation of personalized exercise prescriptions. First, zone-based training may be incorporated into a professional athlete's exercise program to improve athletic performance during competition. In addition, training zones may be used for recreational athletes to improve fitness, lose weight, or for high-intensity interval training.[49] Many individuals use a simplified 5-zone model based entirely on heart rate (HR) to guide their training (Table 2). However, sports and exercise cardiologists incorporate CPET to develop personalized training zones according to the individual's physiological response to exercise. Although CPET determines key variables such as VO_{2max}, peak workloads, and ventilatory efficiency, it also identifies critical physiological parameters relevant to exercise training and performance, such as the maximal steady state (MSS).[7] Although a detailed analysis of exercise physiology is beyond the scope of this article, MSS is an important physiological parameter and refers to the work intensity that can be sustained for a prolonged time period without an accumulation of blood lactate.[7,50] For example, marathon runners typically compete at their MSS. Once MSS is established from CPET (zone 3 in Table 2), additional zones can be determined to achieve the desired outcome.[49,51] For example, zone 4 describes an individual's critical power and is the highest intensity that can be sustained without failure and without encroaching on VO_{2max}. Zone 4 is the hardest training zone to quantify. Multiple CPETs are likely necessary to precisely quantify the boundaries of this zone, since the lower boundary (difference between zone 3 and 4) and the upper boundary (difference between zone 4 and zone 5) may only be a few heartbeats. For an athlete, zone 4 is approximately a 10K racing pace. Zone 5 is an individual's $VO_{2max,}$ and for an athlete, approximates a 5K pace—that is, an intensity that can only be sustained for several minutes. At the lower end of the spectrum, zone 1 is a recovery pace. Zone 2 describes low-intensity exercise that is, by definition, sustainable for an indefinite period of time since it falls below MSS. Zone 2 training has gained popularity recently as a method of promoting weight loss, since

Table 2. Training Zones Determined From Cardiopulmonary Exercise Testing

Zone	Description	Application
Simplified 5-zone training model derived from maximum heart rate		
Zone 1	50%–60% MHR, low intensity	Warm-up, cool-down
Zone 2	60%–70% MHR, moderate Intensity	Build endurance, burn fat
Zone 3	70%–80% MHR, moderate-high Intensity	Improve aerobic fitness
Zone 4	80%–90% MHR, high intensity	Threshold training, improve speed
Zone 5	90%–100% MHR, very high intensity	Short-race competition, maximal effort
Training zones derived from CPET		
Zone 1	Recovery, HR below base pace	Warm-up, recover
Zone 2	Base pace, HR 10–20 bpm below MSS	Build endurance
Zone 3	MSS, reference HR zone	Maximum sustainable intensity. Marathon pace
Zone 4	Race pace, 85%–95% MHR	Critical power, approaches VO_{2max}. 10K pace
Zone 5	Maximum effort	Exercise at $VO_{2max.}$ 5K pace

The simplified 5-zone model is based on the maximum heart rate. CPET indicates cardiopulmonary exercise test; HR, heart rate; MHR, maximum heart rate; MSS, maximal steady-state; and $VO2_{max}$, maximal oxygen uptake.

there is a high degree of fatty acid oxidation at this level of exercise.[52] Performance-based assessments through CPET allow for personalized training zones to achieve the goals of individual patients and athletes.

Evaluating Impairments in Exercise Capacity

Exercise intolerance may present in isolation or with exertional symptoms.[1,2,53] Competitive athletes, in particular, may have objective data of reduced functional capacity based on performance during competition or from training logs. Evaluation of exercise intolerance should not occur in isolation but requires collaborative assessment with coaches, athletic trainers, dieticians, internists, sports medicine practitioners, and any other relevant medical or athletic specialists.[1] The assessment needs to include a detailed review of the exercise training program, environmental factors (thermal stress, altitude) that may contribute to symptoms, prior and current performance data, and potential lifestyle changes, including diet and any dietary deficiencies (such as iron deficiency in female athletes), supplements, and any use of performance-enhancing drugs. Objective assessments should include CPET to determine VO_{2max} and peak power output. In addition, particular attention should be paid to hemodynamic data acquired during submaximal levels of effort—that is, exercise intensities at the level of effort where symptoms occur. Finally, these evaluations need to include an assessment for CVD as a potential cause of symptoms or impaired exercise performance.

Clinical assessments and stress tests should be individualized to the patient based on overall fitness and training habits. For example, the Bruce protocol is the most widely used protocol incorporated for exercise tolerance tests, but the test was originally intended to diagnose coronary artery disease in middle-aged men.[54] A Bruce protocol should not be the default test when evaluating athletes (or any population for that matter), particularly when the goal is to identify the etiology of exertional symptoms and impaired exercise tolerance.[55] The protocol incorporated into exercise assessments should replicate the type of exercise that reproduces patient symptoms.[55] When the cause of impaired exercise capacity remains unclear despite traditional CPET, invasive hemodynamic assessments may be necessary to better characterize cardiac and pulmonary vascular responses to exercise.[56,57]

Wearable devices with hemodynamic monitoring capabilities, and metabolic analyzers (eg, lactate, glucose monitors) allow for acquisition of data to understand hemodynamic responses to exercise and environmental stressors—in essence, real-world scenarios outside of a clinical stress lab or when lying supine on an examination table. These devices can be reliably incorporated into clinical evaluations to assist with identifying the cause of symptoms and impairments in functional capacity.

Return-to-Play

In some instances, cardiovascular evaluations need to include performance-based assessments before RTP.[35] For example, following a viral illness (eg, SARS-CoV-2), CPET should be included to evaluate cardiopulmonary symptoms that persist despite normal triad testing (cardiac enzymes, ECG, and echocardiogram) and cardiac magnetic resonance imaging.[58,59] Athletes with myocarditis and reduced left ventricular function may require exercise testing to rule out clinically relevant arrhythmias before they resume competitive sports.[35] Athletes with ventricular arrhythmias may require an implantable cardiac-defibrillator according to standard recommendations.[35] In these cases, the evaluation for RTP may include exercising testing to monitor for inducible arrhythmias, as well as over- and undersensing, so that implantable cardiac-defibrillator programs can be individualized to the athlete.[1]

EXERCISE CONSIDERATIONS IN NONATHLETIC POPULATIONS

General Population

It is well established that traditional risk factors (eg, hypertension, diabetes, smoking) account for the majority of CVD.[60] Current risk prediction models, such as the PREVENT (Predicting Risk of CVD Events) calculator,[60] incorporate modifiable and nonmodifiable risk factors to determine long-term risk of CVD. However, risk calculators like PREVENT do not incorporate metrics of cardiorespiratory fitness into their models since the risk associated with low fitness is mediated, at least in part, by traditional CVD risk factors.[60] That said, the American Heart Association recommends that fitness be assessed when establishing or reclassifying the risk of CVD.[61] It is well established that fitness is associated with overall survival,[62] heart disease,[61–65] cancer,[66] stroke,[3,67] dementia and cognitive decline,[68] and diabetes,[69] which are the no. 1, 2, 4, 6, and 7 most common causes of death in the United States, respectively.[70]

These data emphasize several important points related to sports and exercise cardiology. First, cardiorespiratory fitness is a central aspect of longevity, primary and secondary prevention of CVD and other major comorbidities that contribute to mortality. Cardiorespiratory fitness is improved/maintained by routine exercise. Thus, sports and exercise cardiologists are essential for improving health through patient education, advisory roles on an individual and societal level, and for public health advocacy.[1,71] Exercise physiology and training are essential components of preventive cardiology programs.[71] In this regard, sports and exercise cardiology naturally intersects with preventive cardiology as it relates to improving survival, cardiovascular health, and reducing risk of other major comorbidities through exercise.

Interestingly, athletes and committed exercisers may have increased coronary artery calcium (CAC) scores.[72–74] The mechanism(s) accounting for this paradox[74] between routine exercise and elevated CAC are not fully understood. However, several mechanisms have been proposed, including hemodynamic stress and endothelial dysfunction, oxidative stress and inflammation, and exercise-induced increases in parathyroid hormone.[74] Emerging data indicate that male athletes are at greater risk of elevated CAC than females.[72–74] Data from the Cooper Center Longitudinal study suggest a dose-response relationship between exercise volume and CAC score.[75] More specifically, in a large analysis (N=21 758 male participants, average 51±8 years old), the risk of CAC score ≥100 Agatson units (AU) was greater among individuals exercising ≥3000 MET(metabolic equivalent)-min/wk compared with those exercising <1500 MET-min/wk (risk ratio, 1.11 [95% CI, 1.03–1.20]).[75] In that analysis, elevated CAC was not associated with an increased risk of all-cause or CVD mortality over a decade of follow-up.[75] However, in a follow-up analysis by the same group, the presence of CAC was associated with an increased risk of clinical CVD.[76] More specifically, a CAC score ≥100 AU was associated with a higher cumulative incidence of acute myocardial infarction compared with a CAC <100 for low-volume exercisers (<500 MET-min/wk) and high-volume exercisers (≥3000 MET-min/wk).[76] Finally, the risk of clinical CVD events was lowest among intermediate levels of exercise (500–3000 MET-min/wk).[76] Together, these data suggest that there is a limit to the benefit of physical activity for the reduction of CVD at levels exceeding 3000 MET-min/wk.[76] These data can be useful as a guide for counseling patients on using exercise for preventive therapies.[76]

Exercise as a Prescription for the General Population

An exercise prescription for any athlete, professional or recreational, is highly dependent on the goals of the individual. As previously noted, individualized exercise prescriptions can be developed based on the athlete's physiological response to exercise during CPET to improve performance. However, there are multiple other reasons to develop exercise prescriptions for nonprofessional athletes, nonathletic populations, and individuals with heritable conditions[77,78]—for example, to optimize longevity,[61] improve general fitness and promote cardiovascular health,[49] and for primary or secondary prevention of atherosclerotic disease.[79–81]

Several major health organizations, including the American Heart Association, American Stroke Association,[79] American Cancer Society,[82] and the Department of Health and Human Services[83] share similar recommendations related to the optimal dose of exercise. These health agencies all generally recommend participation in at least 150 minutes of moderate intensity exercise per week, or at least 75 minutes of vigorous intensity exercise. Physical activity may be monitored by tracking daily step counts with wearable devices. In a landmark analysis of ≈17 000 women (≈72 years of age), walking 4400 steps/d reduced the risk of mortality over 4.3 years of follow-up, and a dose-dependent relationship was observed between step count and mortality risk up to a threshold of 7500 steps per day.[84] In a recent systematic review and meta-analysis of 57 studies, all-cause mortality, CVD, and dementia were all inversely related to daily step count, with an inflection point around 5 to 7000 steps per day.[85] Compared with 2000 steps per day, walking 7000 steps per day was associated with ≈50% lower risk of all-cause mortality, ≈50% lower risk of CVD mortality, ≈40% reduction in cancer mortality, and a ≈40% reduction in dementia.[85]

Routine exercise throughout the adult lifespan also preserves ventricular compliance and prevents age-related reductions in distensibility.[86] In an elegant hemodynamic study, healthy seniors (N=102) were stratified according to patterns of exercise training throughout adulthood.[86] Left ventricular compliance curves, generated through a combination of pulmonary arterial catheterization and echocardiography, demonstrated that LV stiffness constants were lowest among individuals who exercised at least 4 to 5× per week, whereas individuals who exercised only 2 to 3× per week had evidence of increased LV stiffness.[86] Those who exercised at least 4 to 5× per week also had a higher VO_{2max}, higher cardiac output during exercise, and a lower HR and blood pressure for any level of work.[87] It is well known that LV compliance and distensibility are reduced among patients with CVD and heart failure with preserved ejection fraction.[86–88] Thus, exercise has important implications for overall cardiovascular health during normal aging.[86,87] From an exercise physiology perspective, ventricular compliance can be thought of as a modifiable risk factor.[49] For example, among healthy, sedentary middle-aged (N=53±5 years) individuals, 2 years of dynamic exercise training reduced LV stiffness constants and led to a down-rightward shift in the LV end-diastolic pressure volume relationship, consistent with improved LV compliance and distensibility.[49] In another analysis, 1 year of exercise training improved LV myocardial stiffness among patients with stage B heart failure with preserved ejection fraction.[89]

Cardiac Rehabilitation, Exercise, and Health Policy in Patients With CVD

Cardiac rehabilitation (CR) improves survival and health-related quality-of-life in patients with CVD.[81,90] In a systematic review and meta-analysis of 85 trials and 23 430 patients with history of myocardial infarction, coronary artery bypass graft or percutaneous coronary intervention for angina or atherosclerotic cardiovascular disease, exercise-based CR resulted in a reduction in all-cause mortality (risk ratio, 0.87 [95% CI, 0.73–1.04]), a large reduction in myocardial infarction (risk ratio, 0.72 [95% CI, 0.55–0.93]) and a large reduction in all-cause hospitalization (risk ratio, 0.58 [95% CI, 0.43–0.77]).[90] CR involves the creation of an individualized treatment plan that details patient-specific goals that are achieved through a multifaceted approach, including nutritional counseling and weight management, physical activity counseling, and exercise training.[81] Core components of CR, as defined by the Social Security Act[91] are displayed in Table 3.

From a policy standpoint, Medicare Part B provides coverage for CR for patients with a qualifying diagnosis (Table 3), including up to 36 sessions broken down into 2 to 3 sessions per week for 12 to 18 weeks.[92] However, CR utilization rates among eligible patients are low. In a large contemporary analysis (N=143 870) of individuals with a CR-qualifying diagnosis, between 2017 and 2023, only 24% of eligible individuals enrolled in CR, the average number of sessions completed was 14, and <10% of enrolled participants completed all sessions.[93] The COVID-19 pandemic had a modest impact on individuals enrolling (12% fewer patients enrolled in CR in 2020 versus 2017) and completing (13% lower completion rate in 2020 versus 2013) CR.[93] These data emphasize an opportunity for medical professionals—and health systems in general—to support CR and exercise for these patients.[93] Outside of CR, patients with atherosclerotic cardiovascular disease are strongly encouraged to adhere to a routine exercise regimen. More specifically, the American Heart Association has recommended exercise for patients with chronic coronary disease,[94] heart failure[95] and other comorbid conditions, including high blood pressure[96] and dyslipidemia.[97] In addition, the American Stroke Association recommends routine exercise for primary prevention of stroke.[79]

Table 3. Core Components of Cardiac Rehabilitation for Patients With atherosclerotic cardiovascular disease[81,91]

Location	Hospital or physician's office
Provider	Physician or nonphysician practitioner supervision
Individualized treatment plan	Written plan established, reviewed, and signed by a physician every 30 days describing patient-specific goals, patient's progress, and services provided under the plan
Eligibility and qualifying diagnosis	Acute myocardial infarction within the preceding 12 mo
	History of coronary artery bypass grafting surgery
	Stable angina pectoris
	History of heart valve repair or replacement
	History of percutaneous transluminal angioplasty or stenting
	History of heart or heart-lung transplant
	Stable, chronic heart failure with left ventricular ejection fraction ≤35% and New York Heart Association class II–IV symptoms for at least 6 wk
Services provided	Physician-prescribed exercise
	Cardiac risk factor modification
	Psychosocial assessment
	Outcomes assessment

Exercise in Older Adults

The elderly population, defined as age ≥65 years, is the fastest-growing population in the United States.[98] Major comorbid conditions among this age group include CVD, cancer, stroke, Alzheimer disease, and diabetes,[98] which as previously noted, are the no. 1, 2, 4, 6, and 7 most common causes of death, respectively.[70] Importantly, major health organizations, including the American Heart Association,[97] the American Cancer Society,[82] the American Stroke Association,[79] Alzheimer Society,[99] and American Diabetes Association[100] all recommend routine exercise for prevention of these comorbidities. More specifically, these organizations share the same general exercise recommendation—namely, 150 minutes per week of moderate-intensity exercise or 75 minutes per week of vigorous-intensity exercise.

Concern has been raised about the safety and utility of exercise testing in older adults.[101] This concern has been centered around 2 main issues: compared with younger populations, the prevalence of asymptomatic CVD may be higher among elderly populations, and the elderly may have a higher rate of comorbidities/chronic conditions, increasing risk of adverse events during exercise testing and exercise in general.[102] However, elderly individuals wanting to initiate an exercise program may do so following a thorough history and physical examination, combined with stress testing to monitor cardiovascular and hemodynamic responses to exercise.[102] Stress test protocols may need to be modified (eg, using a Naughton versus standard Bruce protocol) for this population to avoid early termination of the study due to difficulty completing an overly aggressive protocol.[102] However, following assessment, elderly individuals may begin exercise training with a low-intensity program that specifically emphasizes balance, gait training, and self-paced walking, with gradual increases in intensity over time.[102]

SPORTS ELECTROPHYSIOLOGY

It is increasingly recognized that EICR may lead to electrical changes that are both physiological and potentially pathophysiologic in the heart of the athlete. Sports Electrophysiology has emerged as a rapidly growing niche within sports and exercise cardiology, focusing on diagnosis and management of arrhythmias in athletes as well as considerations for SCD prevention and shared decision making for RTP. While arrhythmias in athletes may result from traditional CVD risk factors, a nuanced approach is often required for the athletic patient, accounting for arrhythmogenic diseases (if present), the type of sport, and individual goals.[103]

Atrial Fibrillation Ablation in Athletes

It is well established that athletes are at increased risk of developing atrial fibrillation (AF)[104] related to atrial remodeling and fibrosis that may occur with EICR.[105] In addition, hemodynamic stress applied to the atria during endurance exercise, combined with atrial dysfunction during prolonged exercise, may also contribute to the observed increase in AF risk.[106] For the athlete with AF, ablation is recommended as a first-line approach for rhythm control.[103] Pulse field ablation (PFA) has dramatically changed the field of electrophysiology and utilizes high-voltage electric fields to cause irreversible electroporation to create myocardial lesions.[107] Unlike traditional thermal energies, such as radiofrequency or cryoablation, PFA is highly tissue-selective and spares nearby structures like the esophagus and phrenic nerve. Of note, athletes often present with a paroxysmal AF history, and therefore, pulmonary vein isolation should be the goal at the index ablation.[103] PFA is particularly promising due to similar, if not improved efficacy, shorter procedure times, and potentially fewer complications when compared with traditional approaches.[108] Although further research is necessary to determine optimal timing for RTP following PFA in athletes, the advantages of PFA may be particularly attractive, facilitating a shorter return to sport and avoiding the need for anti-arrhythmic drugs.[103]

Stroke Risk Reduction Strategies in Athletes With AF

Most athletes with AF are younger and have low CHA_2DS_2-VASc scores, often not qualifying for anticoagulation per current guidelines. However, if a patient does meet criteria, then anticoagulation is recommended, and a shared decision discussion should be had regarding the risk of bleeding in the context of the athlete's sport.[35,103] Catheter ablation for AF has been shown to reduce the risk of stroke, but discontinuation of anticoagulation post-procedure should be preceded by a careful discussion about both CHA_2DS_2-VASc score and risks not captured by current risk scores, such as left atrial size and fibrosis, and overall AF burden.[103] Left atrial appendage occlusion may be considered in rare cases when anticoagulation is contraindicated, though data in athletes are limited.[103] Alternatively, intermittent anticoagulation has also been proposed for athletes to balance the risks of thrombus formation and bleeding from anticoagulant therapy.[109]

Pacemaker Programming in Athletes

Athletes often have sinus bradycardia, attributed to training adaptation, higher vagal tone, and a higher need for rapid HR response. In a large, observational study of Swedish cross-country skiers (N=209 108) and nonskiers (N=532,290), male skiers had a higher incidence of bradycardia (adjusted hazard ratio, 1.19 [95% CI, 1.05–1.34]) and pacemaker implantation (adjusted hazard ratio, 1.19 [95% CI, 1.04–1.31]) than male nonskiers.[110] In that study, skiers who completed the most races and had the fastest finishing times had the highest incidence of bradycardia and pacemaker placement.[110] Interestingly, among females, the incidence of bradycardia and pacemaker placement did not differ between skiers and nonskiers.[110]

Among athletes who require pacemakers, CPET may be used to improve cardiovascular performance during sport.[111] During monitored exercise testing, device interrogation and reprograming enables optimization of pacemaker settings, improving performance and possibly relieving symptoms when present.[111] Determinants of rate-responsive pacing during exercise include baseline settings, atrioventricular delays, HR response to exercise, adaptive sensors, and type of sport (impact versus non-impact rhythmic locomotion).[111]

TACTICAL ATHLETICS AND ENVIRONMENTAL MEDICINE

Tactical athletes, whose occupation focuses on service to others and whose success impacts that of their team, their community, and the mission,[112] include but are not limited to law enforcement, firefighters, military, pilots, and astronauts. Several major factors differentiate tactical from competitive athletes, including the need to perform at burst-like activity for extended periods of time and without a warm-up period, using heavy survival gear, and while enduring environmental extremes.[112] These and other demands and exposures strain the cardiovascular system in addition to the task at hand. Tactical tasks (eg, pursuit of suspect on foot, alpine search and rescue, flying high-performance aircraft) require varying levels of endurance and strength-based demands from the cardiovascular system.[113–115] Given these demands, tactical organizations (eg, Police Officer Standards and Training, National Fire Protection Agency, and all 5 branches of the US military) have fit for full duty occupational standards, often for both accession and retention. Providers caring for tactical athletes with CVD or heightened risk should be familiar with both the underlying disease and risk and the occupational demands and responsibilities.

Despite organizational baseline physical fitness requirements, CVD remains a leading cause of duty-related death for tactical athletes.[16,115,116] Although the absolute rate of SCA/SCD is low, concerningly, the event rate is much higher during vigorous intensity activities, such as fire suppression for firefighters.[117] Cardiac events can have devastating consequences not only for the tactical athlete but for the community they serve, thus it is paramount to focus efforts on prevention and early detection of CVD. In anticipation of the upcoming American College of Cardiology/American Heart Association Scientific Statement on the cardiovascular care of the tactical athlete, there has been a call to action to further scientific advances and update organizational policies.[112,118]

When cardiovascular risk or disease is identified, initial evaluation and treatment should follow currently accepted guidelines. In comparison to shared decision-making with the competitive athlete,[119] fit for full duty assessments for the tactical athlete must also account for potential resource limitations (eg, access to emergency care, including medications and AED) in austere environments and the overall safety of tactical team members. In military pilots and ground-based operators (eg, soldiers, combat controllers, and Navy SEALS), performance decrement or sudden incapacitation from a cardiovascular cause or treatment jeopardizes not only the tactical athlete but also that of others (eg, airplane passengers/cargo, fellow comrades, surrounding environment). In the military, for example, any mission failure can have dire consequences, costing many lives. Austere environments may include prolonged environmental exposures (eg, heat, cold, high altitude, deep-sea), limited access to medical care, or require prolonged use of body armor or chemical protection. Tactical athletes on full anticoagulation and those with elevated arrhythmogenic risk are two examples where risks and benefits need to be carefully weighed before deployments or training in different environments. Finally, the preferences of a tactical athlete can be addressed in a medical waiver, but the tactical organization has ultimate authority in deciding the fit for full duty assessment. Thus, shared decision-making applied to tactical athletes is highly complex, incorporating the interests, well-being and safety of all personnel whose livelihoods depend on the cardiovascular health of the athlete.[118]

FUTURE DIRECTIONS AND KNOWLEDGE GAPS

From a policy standpoint, future legislative efforts should be focused on initiatives to improve SCA recognition (through education), CPR training, AED access and distribution (including training areas, parks, and club sports), EAP education, and mandates for SCA/SCD reporting among young competitive athletes in the United States. Further research is needed to understand disparities in outcomes following SCA in athletes and the costs and benefits of different PPCS methods. Future research should focus on the intersection of genetic variants in athletes to determine whether certain athletes are at risk of developing an overt cardiomyopathy based on an underlying genetic profile and whether or not these genetic variants increase the risk of adverse events among athletes. Although there are strong data to reinforce the central role of exercise training throughout the adult lifespan, further research is necessary to define the ideal dose of exercise, including type and duration, for any individual, based on an individual's overall health profile and goals. As sports electrophysiology continues to evolve, future research should focus on the incorporation of wearables and artificial-intelligence technology, and how these devices may facilitate the detection and management of arrhythmias in athletes. In addition, further research is necessary to understand ethnic and sex-specific differences of arrhythmias in athletes and best practices for management to optimize performance and reduce the risk of adverse events related to arrhythmias, particularly for AF and stroke.

CONCLUSIONS

While comprehensive care of the athlete requires a multidisciplinary team, cardiologists with experience in sports and exercise science provide expertise not provided by other specialists. In recent years, there has been increased recognition of the complex interaction between sports, exercise, and cardiovascular health. Issues such as preparticipation screening, risk stratification for SCA/SCD, and distinguishing EICR from cardiomyopathies require assessment by cardiologists familiarized with sports and exercise science. Exercise training may be used by individuals across the spectrum of athletic performance to improve fitness, maintain health, and reduce the risk of CVD. Sports and exercise cardiologists serve a critical role in developing exercise training programs to meet the goals of individual patients and athletes. As the field of sports cardiology evolves, it is critical for managing providers to understand the unique aspects of cardiovascular care of athletes, regardless of the level or type of competition.

ARTICLE INFORMATION

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